

Supplementary information for: Biomodelling.jl 2.0

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S1. Stationary laws used for validation

Birth-death. For $\emptyset \xrightarrow{k} X \xrightarrow{\gamma} \emptyset$ the stationary distribution is Poisson with mean k/γ .

Telegraph model. With activation k_{on} , inactivation k_{off} , transcription k_{tx} from the active state and degradation γ , and $a = k_{\text{on}}/\gamma$, $b = k_{\text{off}}/\gamma$, $\lambda = k_{\text{tx}}/\gamma$, the stationary distribution is the Beta-Poisson mixture

$$P(n) = \frac{\lambda^n \Gamma(a+n) \Gamma(a+b)}{n! \Gamma(a+b+n) \Gamma(a)} {}_1F_1(a+n; a+b+n; -\lambda),$$

whose mean is $\lambda a/(a+b)$ and whose variance is $\mu + \lambda^2 ab/[(a+b)^2(a+b+1)]$. We evaluate ${}_1F_1(\alpha; \beta; -\lambda)$ through Kummer's transformation $e^{-\lambda} {}_1F_1(\beta - \alpha; \beta; \lambda)$, whose series has positive terms; partial sums are rescaled by 10^{-200} whenever they exceed 10^{200} so that the log of the sum is accumulated without overflow.

Bursty protein. Protein produced in geometric bursts of mean size b arriving at rate a and degraded at rate γ is negative binomial with shape a/γ and success probability $1/(1+b)$. The package emulates bursts through an mRNA of lifetime $1/(200\gamma)$ translated at rate $200\gamma b$.

S2. Kernel details and correctness tests

The direct method draws the waiting time from $\text{Exp}(a_0)$ and the channel by linear search over the cumulative propensities; after each event only the propensities of reactions that depend on a changed species (the dependency graph) are recomputed, and the running total is refreshed every 512 events. Fixed-step tau-leaping draws $\text{Poisson}(a_j \tau)$ firings for every channel; the strict variant errors when a species would become negative, the hybrid variant rejects the leap and simulates the interval with the direct method. Adaptive tau-leaping classifies as critical every channel within $n_c = 10$ firings of exhausting a reactant, selects τ from the non-critical channels through the bounds $\max(\varepsilon x_i/g_i, 1)/|\mu_i|$ and $\max(\varepsilon x_i/g_i, 1)^2/\sigma_i^2$, draws the time to the next critical event from $\text{Exp}(a_0^{\text{crit}})$, fires at most one critical channel, halves τ on rejection, and performs 100 exact events whenever $\tau < 10/a_0$.

Table S1 lists the tests of the package test suite that validate the kernels: Kolmogorov-Smirnov distances below 0.015 (direct) or 0.02 (approximate kernels) to the Poisson, Beta-Poisson and negative-binomial laws with 8 000 to 12 000 cells; agreement of the adaptive kernel with the direct method on a stiff dimerisation network within 5% in the mean; identical results with one and with several threads.

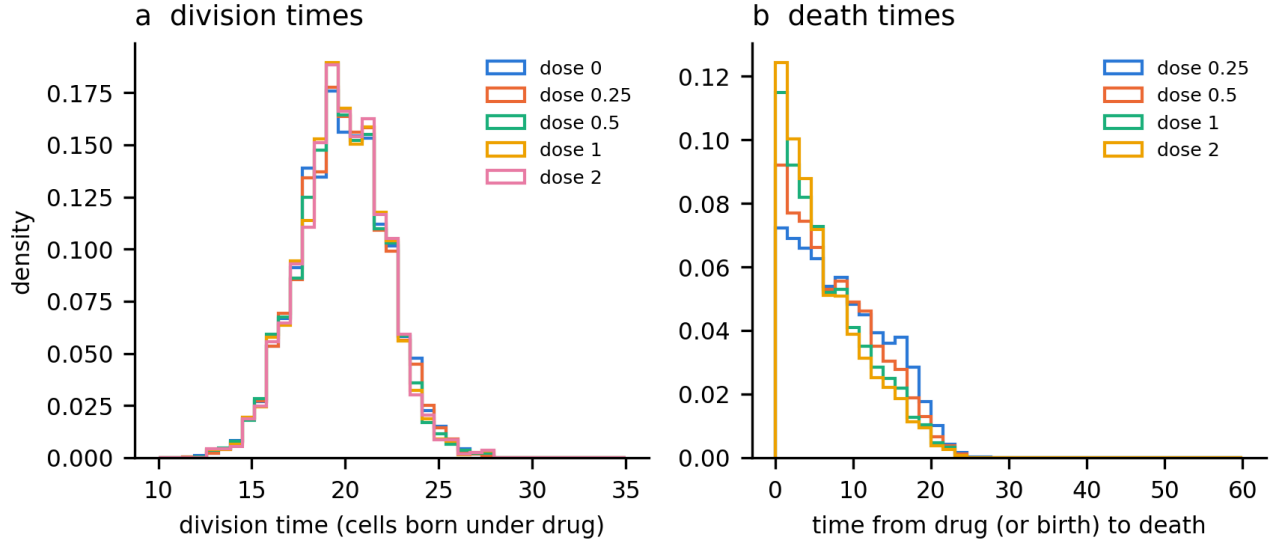
S3. Parameters of all simulations

Figure	Model	Parameters
2a	birth-death	$k = 12, \gamma = 1, T = 15$
2b	telegraph	$(k_{\text{on}}, k_{\text{off}}, k_{\text{tx}}) = (0.4, 0.6, 12)$ and $(0.05, 0.5, 40), \gamma = 1, T = 40$
2c	bursty protein	$a = 1.5, b = 6, \gamma = 1, T = 25$
2d-e	telegraph	$(0.4, 0.6, 12), n = 20\,000$
2f	random telegraph GRNs	$n_{\text{act}} = n_{\text{genes}}, n_{\text{inh}} = n_{\text{genes}}/2, \lambda = \ln 2/20, dt = 0.1, T = 20$
3	telegraph gene with protein	$k_{\text{tx}} = 30, k_{\text{dm}} = 1, k_{\text{tl}} = 4, k_{\text{dp}} = 0.2, k_{\text{on}} = k_{\text{off}}$ as indicated; $\lambda = \ln 2/20$, sizer $V_{\text{div}} = 2$ (cv 0.05), $\sigma_f = 0.02$, replication at 50% of the cycle where indicated
3e	stable protein	$k_{\text{on}} = k_{\text{off}} = 5, k_{\text{tx}} = 20, k_{\text{dm}} = 1, k_{\text{tl}} = 2, k_{\text{dp}} = 0.05$
4	resistance gene	as Figure 3 with $(k_{\text{on}}, k_{\text{off}}) = (0.002, 0.02)$ (memory, $p_{\text{on}} \approx 9\%$) or $(0.05, 0.5)$ (fast, same p_{on}); founders burnt in for 300 time units at constant size; death $h_{\text{max}} = 0.15, \text{EC}_{50} = 0.5, m = 2, K = 150, q = 4$; drug from $t = 40$ of the treatment stage; drug-induced variant $(0.0002, 0.02)$ with $k_{\text{on}} \rightarrow k_{\text{on}}(1 + 100d)$; fitness cost variant: growth $\times (1 - 0.5 c_P^4 / (150^4 + c_P^4))$; memory disruption: switching $\times 20$ on $[0, 40)$ or $[0, 120)$; schedule scan: 300 founders, 240 time units, 20-time-unit exposures
5a	40 independent telegraph genes	$k_{\text{on}} = k_{\text{off}} \in [10^{-3}, 1]$ (log-spaced), $k_{\text{tx}} = 20, k_{\text{dm}} = 0.5$; 40 founders, 6 doublings
5b-c	random GRN	30 genes, 30 activations, 15 inhibitions, $k_{\text{tx}} \in [5, 30], K \in [2, 10], n = 2$, basal 0.05; 1 500 cells; sequencing capture 0.15 (cv 0.3)
5d	random GRN	20 genes, 22 activations, 10 inhibitions; knockdown factor 0.05; 800 cells
6a-b	telegraph	truth $(0.3, 0.6, 20)$; priors log-uniform on $[0.01, 10], [0.01, 10], [1, 200]$; 200 (a) / 120 (b) particles

Figure	Model	Parameters
6c	resistance gene	truth $h_{\max} = 0.5$, $K = 150$; priors log-uniform on $[0.05, 5]$ and $[20, 1000]$; 100 particles
7	resistance gene, hours	cell cycle 24 h (sizer, cv 0.1), $k_{\text{tx}} = 30$, $k_{\text{dm}} = 1$, $k_{\text{tl}} = 4$, $k_{\text{dp}} = 0.2$ per h, $K = 150$, $q = 4$, growth-arrest Hill coefficient 2; fitted: k_{on} , $k_{\text{off}} \in [10^{-3}, 0.3]$, $h_{\max} \in [0.005, 0.5]$ per h, $\text{EC}_{50} \in [3, 40]$ μM , $m \in [1, 6]$, $\text{IC}_{50} \in [1, 40]$ μM ; 300 founders burnt in for 10 cycles, 48 h drug-free, 72 h drug; 160 Latin-hypercube points and 80 Nelder-Mead iterations with seed 1
8a-d	melanoma-like	$k_{\text{tx}} = 3$, $k_{\text{dm}} = 0.1$, $k_{\text{tl}} = 0.4$, $k_{\text{dp}} = 0.02$ per h; net doubling 4 weeks, memory 5 generations, pre-resistant fraction 0.005, $h_{\max} = 0.0023$ per h, $\text{EC}_{50} = 0.3$, $m = 2$, protection $K = 150$, $q = 4$, growth arrest $\text{IC}_{50} = 0.3$ with the same protection, $k_{\text{off}} \rightarrow k_{\text{off}}/(1 + 9d)$, fitness cost 0.5, partial protection: additional unprotected growth inhibition with $\text{IC}_{50} = 1$; 2 000 founders with promoter states drawn from the stationary distribution, 60 weeks, dt 4 h; progression at $1.73 \times$ nadir after the 8-week lead-in

same expression kinetics; net doubling 40 days, memory 4 generations, MGMT-expressing fraction 0.01 or 0.30, $h_{\max} = 0.03$ per h at peak, $EC_{50} = 0.4$ of the standard bolus peak, $m = 2$, $K = 150$, $q = 4$; elimination half-life 2.1 h; MGMT consumption $k_{dp} \rightarrow k_{dp}(1 + 20d)$; 1 500 founders at the stationary promoter distribution, six 28-day cycles, dt 1 h

28 S4. Additional persister simulations



29
30 **Figure S1. Single-cell timing distributions underlying Figure 4c.** (a) Division times of cells born under
31 drug and (b) times from drug start (or from birth, for cells born under drug) to death, per dose.

32 The division-time distribution is set by the sizer and is the same at every dose, because in this model the drug
33 kills but does not slow growth. Times to death are broad (coefficient of variation 0.72 at dose 0.5 and 0.82 at
34 dose 2) and their mean shifts by much less than the population decay rate, because most deaths occur among
35 low-expressing cells whose hazard is already close to $h_{\max}$ at dose 0.5 ($EC_{50} = 0.5$, $m = 2$); raising the dose
36 mainly shortens the survival of cells with intermediate protection. The dose dependence of population decay is
37 therefore carried by the fraction of cells that are protected, not by the kinetics of death of unprotected cells,
38 which is the interpretation Iyer et al. (2025) give of their single-cell tracking data.

39 Under continuous dosing at the highest dose (release period 0, dose 2 in Figure 4f), the long-term growth rate is
40 +0.004 per time unit for pre-existing tolerance, -0.013 when the resistant state carries a 50% growth cost and
41 +0.012 for drug-induced tolerance; the schedules with the lowest long-term growth rate are release period 0, dose
42 2 (growth rate +0.004 per time unit), release period 10, dose 2 (growth rate -0.016 per time unit) and release

period 10, dose 1 (growth rate +0.009 per time unit) respectively. For pre-existing tolerance without a cost, every release period raises the net growth rate; with the fitness cost, release periods of 5 and 10 time units are within 0.003 per time unit of continuous dosing at doses of 1 and 2; for drug-induced tolerance, dose 1 gives a lower net growth rate than dose 2 at every release period, and release periods of 20 time units or more raise the growth rate in every model. All values are single realisations with 300 founder cells.

S5. Benchmark details

Network inference methods: absolute Pearson and Spearman correlations of $\log(1 + x)$ (Pearson) or ranks (Spearman); GENIE3 as random-forest importances (200 trees, `sqr`t features) fitted on $\log(1 + x)$ with scikit-learn. Imputation: kNN-smoothing (one step, $k = 15$, 10 principal components of the Freeman-Tukey transformed, library-size normalised counts) and MAGIC with default parameters on square-root normalised counts. Metrics: area under the precision-recall and receiver-operating curves against the undirected (correlation) or directed (GENIE3) ground-truth adjacency. The cell-cycle-regressed dataset is the residual of $\log(1 + c)$ (concentration c) after ordinary least-squares regression on gene copy number, cell age and age squared, per gene.

S6. Inference diagnostics

Summary statistics and distances. For Figures 6a and 6b the distance between a simulated and the observed sample of counts is their Wasserstein-1 distance (Bernton et al. 2019), computed as the mean absolute difference between the two empirical quantile functions on a grid of 201 probabilities, divided by the observed mean; each simulated sample contains 600 cells (a) or the final snapshot of 500 founders grown for 80 time units (b), against 2 000 observed cells. The distance between two independent samples at the true parameters (the noise floor) is reported in the summary tables of the repository. For Figure 6c the summaries of a treated population are the log surviving fraction on a grid of 25 time points from the start of the drug to 60 time units later (populations that went extinct contribute a floor of 10^{-3}) and the concordance of death fates between sisters born in the 20 time units before the drug; the distance is the root-mean-square difference of these summaries. Each generation's tolerance is the median of the previous generation's accepted distances, particles are perturbed with a Gaussian kernel of variance twice the weighted variance of the previous generation, and the first generation samples the prior.

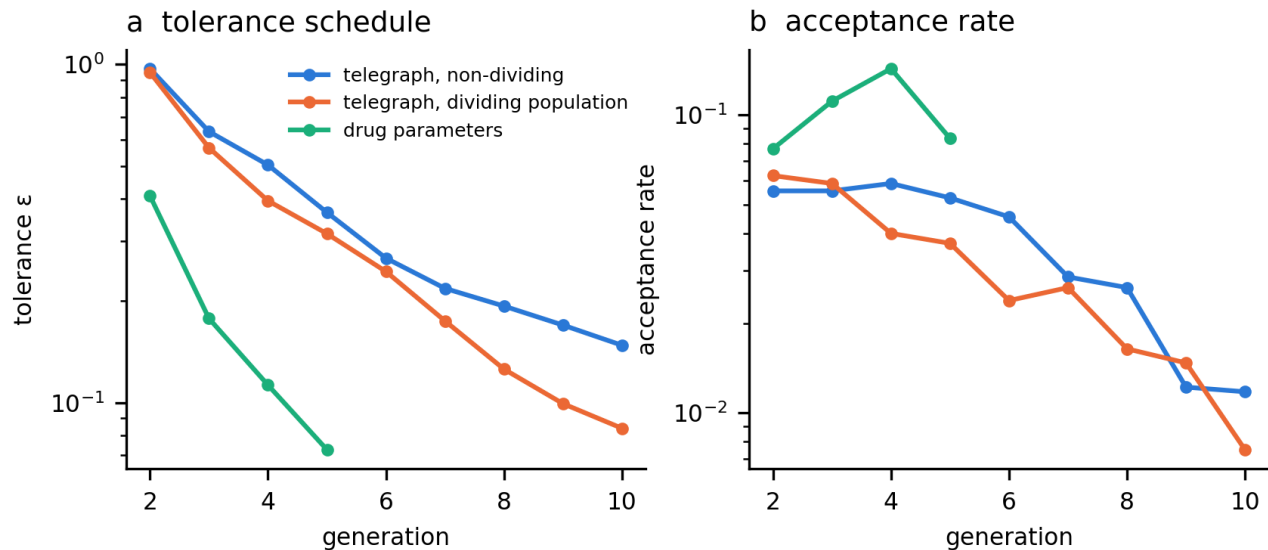


Figure S2. ABC-SMC diagnostics. (a) Tolerance and (b) acceptance rate per generation for the three runs of Figure 6.

The run on non-dividing cells (10 generations, 200 particles) reached a final tolerance of 0.148 with an acceptance rate of 1% in the last generation; the division-aware run (10 generations, 120 particles, each particle simulating a population of 250 founders for 80 time units) reached 0.0841 at 1%; the drug-parameter run (5 generations, 100 particles, 200 founders each) reached 0.0727 at 8%. Posterior predictive checks were made by simulating the forward model at the posterior median (Figure 6d for the drug parameters); the posterior median rather than the mean is reported throughout because the posteriors are asymmetric on the log scale.

S7. Runtime

Table S2 lists the wall-clock times of the population simulations of Figure 2f (200 steps of $dt = 0.1$ for random telegraph networks of 10, 50 and 100 genes, four threads of an Intel Xeon processor at 2.1 GHz, Julia 1.10). Throughput in cell-steps per second falls with the number of reactions because every step of every cell has to sample the events of all channels; the hybrid kernel is 2 to 5 times faster than the direct method at these tolerances, the gap widening with the number of reactions. The single-cell interface used for the comparison with JumpProcesses.jl (Figure 2e) allocates a workspace per simulation and is slower in serial execution than JumpProcesses.jl on that one-gene model; the population interface reuses one workspace per thread.

Table S2. Runtime of population simulations.

genes	cells	kernel	threads	seconds (200 steps)	cell-steps / s
10	100	DirectSSA	4	0.38	52,398
10	100	HybridSSATau	4	0.11	180,019
10	1000	DirectSSA	4	1.04	193,080
10	1000	HybridSSATau	4	0.58	346,004
10	5000	DirectSSA	4	4.78	209,388
10	5000	HybridSSATau	4	2.57	388,683

genes	cells	kernel	threads	seconds (200 steps)	cell-steps / s
50	1000	DirectSSA	4	9.72	20,581
50	1000	HybridSSATau	4	2.57	77,671
100	1000	DirectSSA	4	31.41	6,367
100	1000	HybridSSATau	4	6.37	31,373

S8. Laboratory and clinical reference data

Table S3. Reference values used for calibration and validation (transcribed from the cited articles; files in `paper/data/`).

Quantity	Value	Source
U2OS cells tracked at cisplatin addition (7 / 10 / 13 μ M)	232 / 240 / 296	Iyer et al. (2025), Table 2
of which died within 72 h	30 / 66 / 176	same
of which divided within 72 h	100 / 75 / 67	same
of which survived without dividing	102 / 99 / 53	same
HCT116 cells at drug addition; death fraction	275; 0.64	Iyer et al. (2025), Table 1
Plateau of the HCT116 kill curve	after $\sim$ 100 h	Iyer et al. (2025), Fig 1b
Lineage correlations of fate	present for sisters, first and second cousins; absent for third cousins	Iyer et al. (2025), Fig 5a
Pre-resistant melanoma cells	1:50 to 1:500 per marker; EGFR-high cells give 7.9 ± 0.9 fold more resistant colonies	Shaffer et al. (2017)
N15-0385 glioblastoma doubling time	50 h	Lasri et al. (2020)
Temozolomide elimination half-life	2.1 h	Ostermann et al. (2004)
RTOG 0525 regimens	150-200 mg/m ² days 1-5 vs 75-100 mg/m ² days 1-21 of 28-day cycles; median OS 16.6 vs 14.9 months	Gilbert et al. (2013)
SWOG S1320 regimens	continuous vs 3 weeks off / 5 weeks on after an 8-week lead-in; median PFS 9.0 vs 5.5 months	Algazi et al. (2020)

S9. Migration from version 1

The v1 API (`Donne`, `ssa`, `tauleap`, `tauleapswitch`, `adaptive_tauleap`, `exponential_growth`, `NamedTuple` reactions) remains available through a deprecated compatibility layer that converts `NamedTuple` reactions to `ReactionModels` (reactions whose names contain `act`, `inhib`, `comb_a` or `comb_i` become Hill kinetics with `rate` = `[k, n, K]`; species whose names contain `on/off` become promoter groups) and maps the algorithms onto the new kernels. Output shapes are preserved, including the leading `:NULL` column. See `docs/src/migration.md`.

- 96 Algazi, Alain P., Megan Othus, Adil I. Daud, et al. 2020. “Continuous Versus Intermittent BRAF and MEK
97 Inhibition in Patients with BRAF-Mutated Melanoma: A Randomized Phase 2 Trial.” *Nature Medicine* 26
98 (10): 1564–68. <https://doi.org/10.1038/s41591-020-1060-8>.
- 99 Bernton, Espen, Pierre E. Jacob, Mathieu Gerber, and Christian P. Robert. 2019. “Approximate Bayesian
100 Computation with the Wasserstein Distance.” *Journal of the Royal Statistical Society: Series B* 81 (2): 235–69.
101 <https://doi.org/10.1111/rssb.12312>.
- 102 Gilbert, Mark R., Meihua Wang, Kenneth D. Aldape, et al. 2013. “Dose-Dense Temozolomide for Newly
103 Diagnosed Glioblastoma: A Randomized Phase III Clinical Trial.” *Journal of Clinical Oncology* 31 (32):
104 4085–91. <https://doi.org/10.1200/JCO.2013.49.6968>.
- 105 Iyer, Anton, Adrian Alva, Adrián E. Granada, and Shaon Chakrabarti. 2025. “Inheritable Cell-States Shape
106 Drug-Persister Correlations and Population Dynamics in Cancer Cells.” *PLoS Computational Biology* 21 (9):
107 e1013446. <https://doi.org/10.1371/journal.pcbi.1013446>.
- 108 Lasri, Ayoub, Viktorija Juric, Maité Verreault, et al. 2020. “Phenotypic Selection Through Cell Death: Stochastic
109 Modelling of O-6-Methylguanine-DNA Methyltransferase Dynamics.” *Royal Society Open Science* 7 (7): 191243.
110 <https://doi.org/10.1098/rsos.191243>.
- 111 Ostermann, Sandrine, Chantal Csajka, Thierry Buclin, et al. 2004. “Plasma and Cerebrospinal Fluid Population
112 Pharmacokinetics of Temozolomide in Malignant Glioma Patients.” *Clinical Cancer Research* 10 (11): 3728–36.
113 <https://doi.org/10.1158/1078-0432.CCR-03-0807>.
- 114 Shaffer, Sydney M., Margaret C. Dunagin, Stefan R. Torborg, et al. 2017. “Rare Cell Variability and Drug-Induced
115 Reprogramming as a Mode of Cancer Drug Resistance.” *Nature* 546: 431–35. <https://doi.org/10.1038/nature>
116 22794.